Complete manuscript comparison

Overview v13: expanded projective and mixes topics

The current comparison contains 19 topics and 46 recovered originals among 106 accessible originals; 60 remain pending and 55 earlier indexed originals remain archived. E now covers five originals, including L056 and L043, and Mixes covers F005 and L016. The other 17 topic records and 39 individual records retain their exact v12 boundaries, including the three symmetric listing corrections. Unified A and F007 are outside this edition.

The complete written factor is $\sum_i c(S_i)/(c(U) + \sum_i c(A_i))$ within each topic. Whole common theory and every complete local application are paid, without a cross-topic discount. Each row retains its adopted nonwhitespace-character coordinate. Expanded E uses raw-untrimmed NFC extraction, an explicit change from the earlier E coordinate; this primary was independently fixed and accepted. There is no pooled factor or inferred minimum proof complexity.

Topic	Originals	Sources	Common	Applications	Factor
A	7	213,803	49,059	87,019	1.571
B	5	124,906	44,905	69,265	1.094
C	5	118,433	30,437	61,248	1.292
D	3	47,894	15,632	29,855	1.053
E	5	140,286	57,473	32,337	1.562
F	2	69,825	15,876	27,398	1.614
G	2	73,199	30,928	85,207	0.630
Affine	1	32,767	19,070	22,638	0.786
Representation	1	25,348	11,288	13,688	1.015
Chamber	2	34,177	15,199	21,254	0.938
Matching	1	37,951	16,732	17,168	1.119
ChiralEnumeration	1	27,594	8,831	23,171	0.862
Operations	3	200,630	58,215	100,379	1.265
Gluing	1	27,081	28,416	11,565	0.677
RankContraction	1	15,530	6,937	10,555	0.888
LocallyProjective	2	26,490	25,498	11,966	0.707
MathieuCensus	1	13,736	5,516	35,862	0.332
RankAugmentation	1	33,604	13,904	14,096	1.200
Mixes	2	93,259	24,256	46,202	1.324

The exact 19-topic CSV and 46-original CSV preserve complete component costs, current fractions and historical observations. Factors below one remain visible. The accepted scientific source maps determine what each recovery covers; the numerical consolidation adds no separate proof or question credit.

Individual comparisons: first 23 originals

After acquiring the *whole* common theory and all actual required mathematical, computational and data premises, the conditional factor is $c(S_i)/c(A_i)$. It charges the full local application or complete companion bundle. Optional validation is not silently made a theorem premise.

Original	Topic	Source	Application	Factor
L032	A	16,558	8,212	2.016
L034	A	21,433	5,784	3.706
F016	A	43,983	15,419	2.853
F013	A	25,562	6,224	4.107
L035	A	51,019	14,498	3.519
N009	A	36,824	28,114	1.310
L128	A	18,424	8,768	2.101
N001	B	14,519	7,371	1.970
N003	B	17,780	9,155	1.942
N008	B	18,009	11,132	1.618
L030	B	22,704	12,286	1.848
L040	B	51,894	29,321	1.770
L088	C	13,166	11,311	1.164
N013	C	27,361	15,815	1.730
L108	C	12,849	10,075	1.275
F014	C	20,577	6,962	2.956
L068	C	44,480	17,085	2.603
L066	D	20,916	10,989	1.903
L074	D	20,661	13,367	1.546
L104	D	6,317	5,499	1.149
L079	E	21,212	1,858	11.417
L083	E	20,904	2,922	7.154
L021	E	29,594	14,452	2.048

E pays its whole 20-page common paper, all five applications and all exact source qualifications. The five-source numerator uses 67 source pages. Application routes remain L083 to A1, L079 to A2, L021 to A3, L056 to A4 and L043 to A5. The broader quoted classifications and ancillary censuses remain retained where the source map says so; they are not newly claimed as regenerated results.

L043 names *Magma* and the L_2-quotient algorithm. GAP and Python describe the replacement implementation. Its accepted corrections, L056's characteristic-two qualification, the L021 corrections, and the exclusion of L079's $\mathrm{P\Sigma L}(2, 25)$ clause remain explicit. E's 42 scoped Lean declarations are partial evidence, not full source formalization.

Individual comparisons: remaining 23 originals

Original	Topic	Source	Application	Factor
L056	E	15,871	4,144	3.830
L043	E	52,705	8,961	5.882
L014	F	25,574	11,504	2.223
L007	F	44,251	15,894	2.784
L009	G	49,891	23,544	2.119
L092	G	23,308	61,663	0.378
L012	Affine	32,767	22,638	1.447
F010	Representation	25,348	13,688	1.852
L008	Chamber	20,159	7,571	2.663
N016	Chamber	14,018	13,683	1.024
F015	Matching	37,951	17,168	2.211
L028	ChiralEnumeration	27,594	23,171	1.191
N005	Operations	122,002	49,595	2.460
N006	Operations	55,740	38,479	1.449
N007	Operations	22,888	12,305	1.860
L010	Gluing	27,081	11,565	2.342
L037	RankContraction	15,530	10,555	1.471
L084	LocallyProjective	2,071	4,857	0.426
L095	LocallyProjective	24,419	7,109	3.435
L073	MathieuCensus	13,736	35,862	0.383
L020	RankAugmentation	33,604	14,096	2.384
F005	Mixes	57,753	31,077	1.858
L016	Mixes	35,506	15,125	2.348

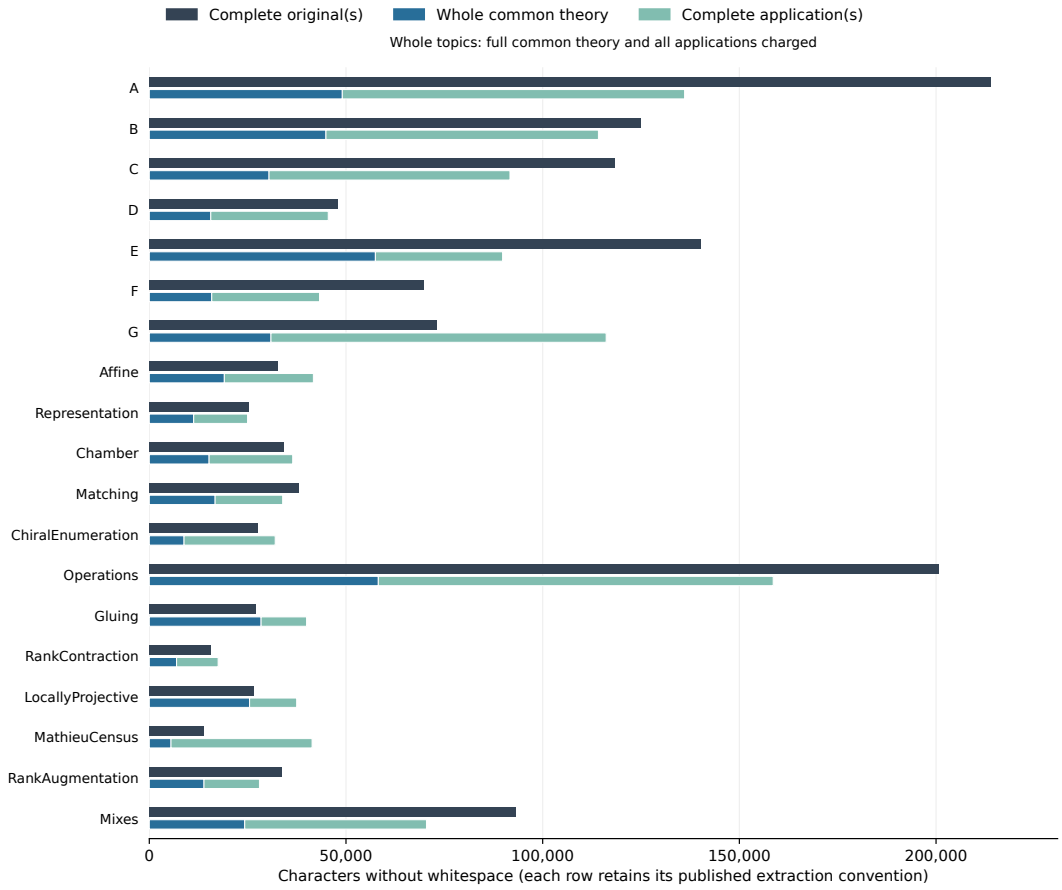
Mixes pays both complete common papers, with costs 12,615 and 11,641. F005's application pays both its main and exceptional papers (20,788 plus 10,289); L016's complete local paper costs 15,125. Both required finite certificates remain required, while four validation programs remain optional. Six scoped formal declarations do not formalize the full sources.

The three final F005 problems remain unanswered. L016's literal thin-cover Lemma 3.5 is corrected under faithfulness; Proposition 5.4 is stated without CIP. Ancillary smallest-example/Atlas census and further-cover questions remain outside the accepted principal recovery. A corrected source theorem is not assigned a new solved-question credit.

Executable listings are excluded symmetrically, while proof and mathematical algorithm prose remain. All required program/data and optional validation inventories are preserved separately. Historical software, external proofs, complete graphics and full formalization costs are unknown; software-inclusive and pooled factors remain null. Operations retains its separate text-only sensitivity. Prior overviews remain historical records.

Whole topics

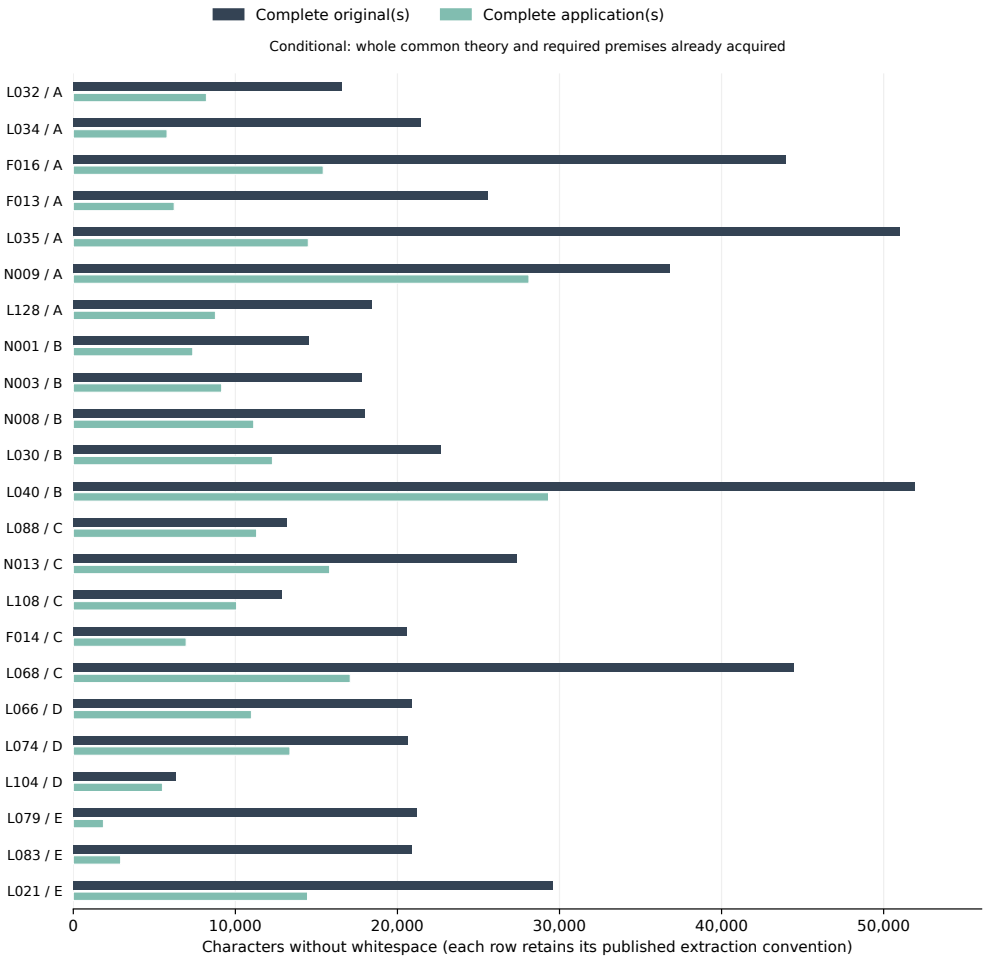
Grey bars show the complete original manuscript text in each row's adopted coordinate. Replacement bars stack the entire common theory and every local application. The axis starts at zero; the tables contain ratios.



The 57 bars account for all 19 complete topics. Shared theory remains charged even when only some applications invoke a section. No required code or external premise is treated as mathematically unnecessary because its cost is outside the manuscript coordinate.

Conditional lengths: first 23 originals

Whole common theory and required premises are acquired in advance. Grey and teal bars compare complete source and local application text; these bars do not remove the prerequisite cost from the whole-topic table.



Conditional lengths: remaining 23 originals

All complete companion papers remain charged. The source sequence is documentary, with no optimized acquisition order, sawtooth or proof-node factor implied.

